Correlation, Regression, Covariance and Outliers

Detailed Notes with Simple Examples

Purpose: These notes explain the statistics chapter in a simple, exam-friendly and interview-friendly way. The focus is on understanding relationships between variables, prediction, and identifying unusual values in data.

Source: Based on the uploaded file Correlation_Regression.pdf.

1. Random Variable

A random variable is a variable whose value depends on the result of a random experiment. It assigns a numerical value to each possible outcome.

Example: If you toss a coin, you can assign Head = 1 and Tail = 0. If you roll a dice, the random variable can take values 1, 2, 3, 4, 5 or 6.

1.1 Discrete Random Variable

A discrete random variable has countable values. These values are usually whole numbers.

Situation	Possible values
Number of heads in 10 coin tosses	0, 1, 2, ..., 10
Number of students in a class	1, 2, 3, ...
Number of calls received per hour	0, 1, 2, 3, ...

1.2 Continuous Random Variable

A continuous random variable can take any value in a range or interval. It can include decimal values, so there are infinitely many possible values.

Situation	Example values
Height of a person	165.2 cm, 165.25 cm, 165.255 cm
Temperature of a patient	98.4°F, 99.1°F
Speed of a vehicle	80.5 km/hr, 90.75 km/hr

Discrete variable	Continuous variable
Countable values	Uncountable values in an interval
Usually whole numbers	Can have decimals
Uses PMF - Probability Mass Function	Uses PDF - Probability Density Function

2. Correlation

Correlation measures how strongly two variables are related and in which direction they move. It answers the question: Do two variables move together?

Example: If study hours increase and marks also increase, study hours and marks have positive correlation.

Important: Correlation does not imply causation. This means that even if two things are related, one does not always directly cause the other.

2.1 Correlation Coefficient

Correlation is usually represented by r. The value of r always lies between -1 and +1.

$$-1 \leq r \leq +1$$

Value of r	Meaning
+1	Perfect positive correlation
-1	Perfect negative correlation
0	No linear correlation
Close to +1	Strong positive relationship
Close to -1	Strong negative relationship
Close to 0	Weak or no linear relationship

3. Types of Correlation

3.1 Positive Correlation

When both variables move in the same direction, it is called positive correlation. If X increases, Y also increases. If X decreases, Y also decreases.

X: Study hours	Y: Marks
1	35
2	45
3	60
4	75

Here, as study hours increase, marks also increase. Therefore, r is positive.

3.2 Negative Correlation

When one variable increases and the other decreases, it is called negative correlation.

X: Price	Y: Demand
10	100
20	80
30	60

40	40

Here, as price increases, demand decreases. Therefore, r is negative.

3.3 Zero Correlation

When there is no meaningful relationship between two variables, it is called zero correlation. Example: shoe size and intelligence.

4. Linear and Curvilinear Correlation

Linear correlation means the relationship can be shown using a straight line. Example: Y = 2X.

Curvilinear correlation means the relationship forms a curve. In this case, the amount of change in one variable is not constant compared with the other variable.

Type	Meaning	Example
Linear	Straight-line relationship	Study hours and marks
Curvilinear	Curved relationship	Experience and salary after a saturation point

5. Simple, Multiple and Partial Correlation

Type	Meaning	Example
Simple correlation	Relationship between two variables	Study hours and marks
Multiple correlation	Relationship among three or more variables	Marks depending on study hours, sleep and attendance
Partial correlation	Relationship between two variables while keeping other variables constant	Study hours vs marks while keeping sleep constant

6. Scatter Diagram

A scatter diagram is a graph of dots. Each dot represents one pair of values (X, Y). It is the simplest graphical method to see whether two variables are related.

Dot pattern	Interpretation
Dots go upward from left to right	Positive correlation
Dots go downward from left to right	Negative correlation
Dots are close to a straight line	Strong correlation
Dots are widely scattered	Weak correlation
No visible pattern	No correlation

Advantage: It is simple and visual. Limitation: It does not give the exact numerical value of correlation.

7. Karl Pearson’s Correlation Coefficient

Karl Pearson’s coefficient is a mathematical method for measuring the strength and direction of a linear relationship between two variables.

$$r = Cov(X,Y) / (\sigma X \times \sigma Y)$$

Symbol	Meaning
r	Pearson correlation coefficient
Cov(X,Y)	Covariance between X and Y
σX	Standard deviation of X
σY	Standard deviation of Y

Pearson correlation is unit-free. It standardizes covariance by dividing it by the product of the standard deviations.

7.1 Interpretation of Pearson r

Range	Interpretation
+0.75 to +1	Strong positive correlation
+0.25 to +0.75	Moderate positive correlation
0 to +0.25	Weak positive correlation
0	No correlation
-0.25 to 0	Weak negative correlation
-0.75 to -0.25	Moderate negative correlation
-1 to -0.75	Strong negative correlation

7.2 Limitations of Pearson Correlation

- It assumes a linear relationship.
- It can be affected by extreme values or outliers.
- It measures association, not causation.
- Interpreting the value of r can sometimes be difficult without context.

8. Coefficient of Determination

The coefficient of determination is the square of the correlation coefficient. It tells how much variation in the dependent variable is explained by the independent variable.

$$R^2 = r^2$$

Example: If $r = 0.9$, then $R^2 = 0.81$. This means 81% of the variation in Y is explained by X. The remaining 19% may be due to other factors.

If $r = 0.60$, $R^2 = 0.36$, so 36% of variation is explained. If $r = 0.30$, $R^2 = 0.09$, so only 9% of variation is explained.

9. Spearman's Rank Correlation

Spearman rank correlation is used when data is in ranks or order. It measures the strength and direction of a monotonic relationship.

Use Spearman correlation when:

- Data is ordinal or ranked.
- Actual values are not as important as order.
- Pearson assumptions like linearity or normality are not satisfied.

$$R = 1 - [6\Sigma d^2 / n(n^2 - 1)]$$

Symbol	Meaning
R	Spearman rank correlation coefficient
d	Difference between the two ranks
d^2	Square of rank difference
n	Number of observations

9.1 Spearman Example

Student	Math rank	Physics rank	d	d^2
A	1	1	0	0
B	2	3	-1	1
C	3	2	1	1
D	4	4	0	0
E	5	5	0	0

Here, $\Sigma d^2 = 2$ and $n = 5$.

$$R = 1 - [6(2) / 5(5^2 - 1)] = 1 - 12/120 = 0.9$$

So, there is a strong positive rank correlation. Students who rank high in Maths also rank high in Physics.

10. Regression Analysis

Regression analysis is used for prediction. It predicts the value of one dependent variable using another independent variable.

Example: Predict marks using study hours. Here, study hours is the independent variable and marks is the dependent variable.

$$Y = a + bX$$

Symbol	Meaning
Y	Dependent variable - value to be predicted

X	Independent variable - predictor
a	Intercept - value of Y when X = 0
b	Slope - change in Y for one unit change in X

10.1 Slope and Intercept

In the equation Y = a + bX, the intercept a tells the value of Y when X is zero. The slope b tells how much Y changes when X increases by one unit.

Example: Marks = 20 + 10 × Study Hours. If a student studies for 5 hours, predicted marks = 20 + 10(5) = 70.

10.2 Two Regression Lines

Regression line	Use
Y on X	Predict Y from X. Example: predict marks from study hours.
X on Y	Predict X from Y. Example: predict study hours from marks.

If correlation is perfect, both regression lines coincide. If the two regression lines are far from each other, the degree of correlation is lower.

11. Correlation vs Regression

Correlation	Regression
Measures relationship	Predicts value
No dependent or independent variable	Has independent and dependent variable
r ranges from -1 to +1	Regression equation can produce values beyond this range
Unit-free	Depends on units of variables
X and Y are treated equally	X is used to predict Y
Does not prove cause and effect	Used for prediction and cause-effect style analysis

Simple memory: Correlation says whether variables are related. Regression gives an equation to predict one variable from another.

12. Standard Error of Estimate

The standard error of estimate measures the average distance between actual values and predicted values from the regression line. It tells how accurate the prediction model is.

A smaller standard error means the predictions are closer to the actual values. A larger standard error means the model is not predicting well.

Example: If actual marks = 80 and predicted marks = 78, error is small. If actual marks = 80 and predicted marks = 50, error is large.

13. Covariance

Covariance tells how two variables move together. It shows the direction of relationship, but not the strength clearly.

$$Cov(X,Y) = E[(X - E[X])(Y - E[Y])]$$

Covariance value	Meaning
$Cov(X,Y) > 0$	X and Y move in the same direction
$Cov(X,Y) < 0$	X and Y move in opposite directions
$Cov(X,Y) = 0$	No linear movement together

13.1 Covariance Example

Study Hours X	Marks Y
2	50
4	60
6	65
8	80

As study hours increase, marks also increase. So covariance is positive.

13.2 Properties of Covariance

- $Cov(X, X) = Var(X)$. Covariance of a variable with itself equals variance.
- $Cov(X, Y) = Cov(Y, X)$. Covariance is symmetric.
- $Cov(X, constant) = 0$.
- If X and Y are independent in many common cases, $Cov(X, Y) = 0$.

14. Covariance vs Correlation

Covariance	Correlation
Shows direction of movement	Shows direction and strength
Can range from $-\infty$ to $+\infty$	Ranges from -1 to +1
Depends on units and scale	Unit-free and standardized
Hard to compare across datasets	Easy to compare
Example: Cov = 31.67	Example: r = 0.85

Correlation is the normalized version of covariance. That is why correlation is easier to interpret than covariance.

$$Correlation(X,Y) = Cov(X,Y) / (\sigma X \times \sigma Y)$$

15. Outliers

An outlier is a value that is unusually high, unusually low, or very different from the general pattern of the data.

Example: In the data 2, 3, 4, 5, 6, 47, the value 47 is an outlier because most values are near 2 to 6, but 47 is far away.

15.1 Why Outliers Matter

Outliers can strongly affect mean, standard deviation, range, correlation and regression results.

Without outlier: data = 2, 3, 4, 5, 6. Mean = 4. With outlier: data = 2, 3, 4, 5, 6, 47. Mean = 11.16. One extreme value changed the mean heavily.

15.2 Methods to Detect Outliers

Method 1: IQR Rule

$$\mathbf{IQR = Q3 - Q1}$$

A value is an outlier if it is below Q1 - 1.5(IQR) or above Q3 + 1.5(IQR).

Method 2: Z-score Rule

$$\mathbf{Z = (X - \mu) / \sigma}$$

If $|Z| > 3$, the value is likely an outlier.

Method 3: Visual Methods

- Box plot: points beyond whiskers are possible outliers.
- Scatter plot: isolated points can be outliers.
- Histogram: far-away bars can indicate outliers.

16. Complete Example Connecting Everything

Suppose we have this data:

Study Hours X	Marks Y
2	50
4	60
6	65
8	80

Step 1 - Covariance: As study hours increase, marks increase. So covariance is positive.

Step 2 - Correlation: Since both variables increase together, correlation is positive. If points are close to a straight line, correlation is strong.

Step 3 - Regression: We can create an equation like Marks = a + b(Study Hours) and use it to predict marks.

Step 4 - Outlier check: If one record is Study Hours = 1 and Marks = 100, it may be an outlier and can disturb correlation and regression.

17. Interview-Level Summary

Concept	Main question answered	Simple meaning
Covariance	Do variables move together?	Direction
Correlation	How strongly are variables related?	Direction + strength
Regression	Can we predict one variable using another?	Prediction
Outlier	Is there an extreme value?	Unusual observation
R^2	How much variation is explained?	Explained percentage

18. Most Important Formulas

Concept	Formula
Pearson correlation	$r = Cov(X,Y) / (\sigma X \times \sigma Y)$
Coefficient of determination	$R^2 = r^2$
Spearman rank correlation	$R = 1 - [6\Sigma d^2 / n(n^2 - 1)]$
Regression line	$Y = a + bX$
Covariance	$Cov(X,Y) = E[(X - E[X])(Y - E[Y])]$
Z-score	$Z = (X - \mu) / \sigma$
IQR	$IQR = Q3 - Q1$

19. Final Memory Trick

Covariance = Direction

Correlation = Direction + Strength

Regression = Prediction

Outlier = Extreme value

These topics are important in statistics, machine learning, data science, business analytics and big data interviews because they help you understand patterns, relationships, predictions and unusual values in data.